

do44_classification_probabilistic_ ADDENDUM

ADDENDUM — content to merge into the existing deck, not a standalone presentation.

Addendum for do44_classification_probabilistic.p df

This is not a new deck — it is content to insert into the existing **Model Families — Probabilistic Classifiers** presentation (`do44_classification_probabilistic.pdf`, currently Naive Bayes only). The deck's own title already frames it as a family, so this addendum adds the second member: multinomial logistic regression.

A Second Probabilistic Family Member —

`cla_multinom`

Multinomial logistic regression for multiclass classification — a probabilistic linear model that naturally handles more than two classes, unlike binary logistic regression.

```
model <- cla_multinom("Species")
model <- fit(model, iris_train)
prediction <- predict(model, iris_test)
evaluate(model, iris_test$Species, prediction)
```

- **Didactic goal:** keep the multiclass classification line of experiment fixed and show a probabilistic linear model that handles more than two classes directly, without one-vs-rest workarounds.
- Same configure → fit → predict → evaluate rhythm as `cla_nb`.
- Contrast point with Naive Bayes: both are probabilistic, but Naive Bayes assumes conditional independence between features, while multinomial regression models the class log-odds as a direct linear combination of the features.

Reference: Hastie, T., Tibshirani, R., and Friedman, J. (2009). *The Elements of Statistical Learning*.

Suggested Placement

Insert as a closing section titled "**A Second Probabilistic Model: Multinomial Regression**" after the existing Naive Bayes content, and retitle the deck's framing slide (if present) to acknowledge two techniques instead of one — the deck already describes itself as "Probabilistic Classifiers" (plural), so this closes the gap between the title and the content.